MILITARY SPECIFICATION

BAROSWITCH STANDARDS: T-4167A and T-4168A

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of baroswitch standard entitled Baroswitch Standard: T-4167A and T-4168A used for comparison and calibration of baroswitch settings used on the Nike Hercules warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedures For.

DRAWINGS

ORDNANCE CORPS

8805943	- Special Adjustment Tool Assembly.
8861429	- Baroswitch Standard: T-4167A.
8861430	- Baroswitch Standard: T-4168A.
8861441	- Carton, Packing, Ammunition, for Baroswitch Standard: T-4167A.
8861442	- Carton, Packing, Ammunition, for Baroswitch Standard: T-4168A.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The baroswitch shall comply with all requirements specified on the drawing (dwg) 8861429 (T-4167A) and 8861430 (T-4168A), with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Closure sensing electrical circuitry. Closure sensing electrical circuitry shall have a potential of 28 plus or minus (+) 2 volts direct current (vdc). The current shall not exceed 300 milliamperes (ma) after closure. Current at time of make or break shall not exceed 50 ma non-inductive. Testing shall be as specified in 4.4.2.

3.5 Element loop resistance. The resistance of each element (see 6.5.2) shall not exceed 0.5 ohm when tested as specified in 4.4.3.

3.6 Insulation resistance. At a pressure of 1014 millibars (mbs.) when a voltage of 300 ± 15 vdc is applied, the insulation resistance shall be not less than 25 megohms. At a pressure of 187.8 mbs. when a voltage of 250 + 10 vdc is applied, the insulation resistance shall be not less than 5 megohms. Testing shall be as specified in 4.4.4.

3.7 Housing case leakage. The baroswitch unit shall show no leakage beyond 6 mbs. when tested for 5 minutes as specified in 4.4.5.

3.8 Environmental testing. The baroswitch unit shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The baroswitch unit shall also comply with 3.8.1.1 and 3.8.1.2 before and after each environmental test. Testing shall be as specified in 4.4.6.1. Tests shall be conducted in the order specified.

3.8.1 Altitude.

3.8.1.1 Lower. Each baroswitch element shall hold an operating point of 977.12 ± 4 mbs. at three operating temperature ranges when tested as specified in 4.4.6.1.1.

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3.8.1.2 Upper. Each baroswitch element shall hold an operating point of 595.0 ± 4 mbs. at three operating temperature ranges when tested as specified in 4.4.6.1.2.

3.8.2 Stability. Each baroswitch element shall be capable of holding an initial operating point of 842.4 ± 4 mbs. for 14 days when tested as specified in 4.4.6.2.

3.8.3 Transportation vibration. The baroswitch unit shall show no evidence of deterioration such as cracking or breaking when vibrated through the frequency ranges at the amplitudes specified in 4.4.6.3. Following this test, the baroswitch unit shall comply with the requirements of 3.8.1.1 and 3.8.1.2.

3.8.4 Shock. The baroswitch unit shall be capable of withstanding, without evidence of damage such as breaking or cracking, shocks of 15 gravity units (g's) on each of its six sides, when applied for a maximum pulse time of 11 ± 1 milliseconds. Testing shall be as specified in 4.4.6.4. Following this test, the baroswitch unit shall comply with the requirements of 3.8.1.1 and 3.8.1.2.

3.8.5 High temperature. The baroswitch unit shall withstand exposure to a temperature of 165 degrees Fahrenheit (°F.) for 24 hours without evidence of damage when tested as specified in 4.4.6.5. Following this test, the baroswitch unit shall comply with the requirements of 3.8.1.1 and 3.8.1.2.

3.8.6 Low temperature. The baroswitch unit shall withstand exposure to a temperature of -65 ± 5°F for 24 hours without evidence of damage when tested as specified in 4.4.6.6. Following this test, the baroswitch unit shall comply with the requirements of 3.8.1.1 and 3.8.1.2.

3.9 Life cycle. Each element contact shall be capable of making and breaking a 200-milliampere, non-inductive current when subjected to 250 pressure cycles that vary from approximately 115.9 to 1013.9 to 115.9 mbs. at a standard rate of pressure change (see 6.5.3). Testing shall be as specified in 4.4.7. Following this test, the baroswitch unit shall comply with the requirements of 3.5 and 3.6.

3.10 Soldering. All soldering shall comply with the requirements of MIL-STD-1460.

3.11 Workmanship.

3.11.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.11.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.11.3 Burrs. Parts shall be free from burrs.

3.11.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.11.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

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3.11.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8861430 and 8861429) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of Characteristics (see 4.3.2.1)		X
2. Element loop resistance (see 3.5)		X
3. Insulation resistance (see 3.6)		X
4. Housing case leakage (see 3.7)		X
5. Altitude (see 3.8.1)		X
6. Stability (see 3.8.2)		X
7. Transportation vibration (see 3.8.3)	X	
8. Shock (see 3.8.4)		X
9. High temperature (see 3.8.5)		X
10. Low temperature (see 3.8.6)		X
11. Life cycle (see 3.9)		X

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Baroswitch standard, prior to installation of cover and housing (see dwgs 8861429 and 8861430 as applicable).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. "O" ring missing or broken	Visual
	102. Electrical connector improperly installed in mounting plate	Visual
	103. Connector socket damaged	Visual
	104. Symbol "SI" improperly oriented	Visual
	105. Worm and jack screw assembly binds	Visual
	106. Binding posts for electrical contact loose	Visual
	107. Loose solder connections	Visual
	108. Plug assembly loose	Visual
	109. Any component missing or loose	Visual/Manual
	110. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Worm and jack screw assembly not lubricated	Visual
	202. Evidence of poor workmanship (see 3.11)	Visual

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4.3.2.1.2 Baroswitch standard (see dwgs 8861429 and 8861430 as applicable).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Housing or cover cracked	Visual
	102. Cover improperly seated	Visual
	103. Marking missing, misleading or unidentifiable	Visual
	104. Cover loose	Manual
Minor:		
	201. Finish improper	Visual
	202. Evidence of poor workmanship (see 3.11)	Visual

4.3.2.1.3 Carton, packing, ammunition, for baroswitch, standard, prior to closing (see dwgs 8861441 and 8861442, as applicable).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Any cushioning pad missing or improperly positioned	Visual
	102. Baroswitch improperly positioned	Visual
	103. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.4 Carton, packing, ammunition, for baroswitch, standard (see dwgs 8861441 and 8861442 as applicable).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Top or bottom sealing strip improperly positioned	Visual
	102. Marking missing, improper, unidentifiable, or improperly positioned	Visual

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Minor:

201. Evidence of poor workmanship (see 3.11).....Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Element loop resistance	Major
b. Insulation resistance	Major
c. Housing case leakage	Major
d. Altitude (lower and upper)	Major
e. Transportation vibration	Major

Of these tests, (e) is considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Element loop resistance, insulation resistance, housing case leakage and altitude. Each baroswitch unit in the lot shall be tested in accordance with 4.4.3, 4.4.4, 4.4.5 and 4.4.6.1. If any item fails to comply with 3.5, 3.6, 3.7 and 3.8.1, the item shall be rejected and removed from the lot.

4.3.3.2 Transportation vibration. One sample baroswitch unit shall be selected from each 100 units produced until 300 units have been produced. Thereafter, one sample unit shall be selected from each 300 units produced. All samples shall be tested in accordance with 4.4.6.3. If any item fails to comply with the requirements of 3.8.3, the lot shall be rejected.

4.3.4 Acceptance inspection equipment. Inspection and testing shall be performed with the acceptance test equipment, operating instructions and calibration instructions prepared by the contractor. The contractor shall obtain approval of such designs, operating instructions and special calibration instructions as specified in 6.3 prior to use on the contract.

4.3.4.1 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

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4.4.1 Test set-up. In performing pressure tests, the test set-up shown on Figure 1, or a comparable set-up, shall be used. Prior to performing any test, the temperature chamber of the test set-up indicated on Figure 1 shall be stabilized (see 6.4) at the specified temperature and maintained throughout the test period. Prior to testing at a specified setting, each baroswitch unit shall be adjusted to the test setting by performing the following steps sequentially:

a. Adjust F (Figure 1) so that A indicates the desired contact closure reading (see 6.5.3).
b. Adjust the element to be set with a special adjustment tool assembly (dwg 8805943) through B to produce a contact closure indication (see 6.5.4).
c. Perform a closure sensing dive (see 4.4.2 and 6.5.9) and observe contact reading.

NOTE: If the contact closure reading is not observed to be within ± 3.99 mbs. of the specified setting, repeat b and c above in sequence as necessary.

4.4.2 Closure sensing dive. A closure sensing dive is the performance of the following events in sequence:

a. The baroswitch element is subjected to an external pressure of 116.4 ± 6.7 mbs. at standard rate of pressure change (see 6.5.5).
b. The external pressure is increased on the baroswitch element at standard rate of pressure change until a point of 66.6 ± 13.3 mbs. less than the expected pressure point is reached.
c. At this point the increase is continued at a rate exceeding pressure change until the contact closure indication (see 6.5.4) occurs. The pressure change may be stopped for a period of not more than 30 seconds while a contact closure reading is being made.
d. The increase is continued at a standard rate of pressure change until a pressure of 1014 ± 13.6 mbs. is reached.

4.4.3 Element loop resistance. A contact closure reading of 842.4 ± 4 mbs shall be obtained on the baroswitch element and the pressure increased 2.7 ± .3 mbs. The loop resistance shall be determined by the 4 wire-voltmeter-ammeter method when the contact is conducting 200 ± 20 milliamperes in a 28 ± 2 vdc circuit. The above test shall be repeated for each baroswitch element.

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4.4.4 Insulation resistance. The baroswitch unit shall be placed in a pressure chamber with external leads connected to pins 1, 2, 3, and 4 of the baroswitch unit connector. The pressure shall be set at 1014 mbs. and 300 ± 15 vdc shall be applied successively between each pin and common ground by a megger or approved equivalent tester. The insulation resistance shall be not less than 25 megohms. The pressure shall then be set at 187.8 mbs and 250 ± 10 vdc shall be applied successively between each pin and common ground. The insulation resistance shall be not less than 5 megohms.

4.4.5 Housing case leakage. The test set-up of Figure 1, or equivalent, shall be used in conducting this test. An internal pressure of 667 mbs less than the external pressure shall be established. The values on "A" at the beginning and at the end of a 5-minute period shall be determined. The difference between the two readings shall not exceed 6.6 mbs.

4.4.6 Environmental testing (see 3.8).

4.4.6.1 Atlitude.

4.4.6.1.1 Lower. Three closure sensing dives (see 6.5.9) shall be performed on each baroswitch element at an operating point (see 6.5.6) of 977.12 ± 4 mbs at each temperature range specified in Table IV. A total of nine dives shall be performed on each element and the limit criteria requirements of Table IV shall apply.

4.4.6.1.2 Upper. Three closure sensing dives (see 6.5.9) shall be performed on each baroswitch element at an operating point of 595.0 ± 4 mbs at each temperature range specified in Table IV. A total of nine dives shall be performed on each element and the limit criteria requirements of Table IV shall apply.

Table IV

Limit criteria

Maximum repeatability error (see 6.5.7)		
50° ± 3°F	75° ± 3°F	110° ± 3°F
0.6 mbs.	0.6 mbs.	0.6 mbs.
Maximum temperature error (see 6.5.8)		
50° ± 3°F to 75° ± 3°F		130° ± 3°F to 75° ± 3°F
1.3 mbs.		1.3 mbs.

4.4.6.2 Stability. An initial operating point of 842.4 ± 4 mbs shall be established on each element. Without having made any setting adjustments, the operating point of each element shall be determined on the 14th day after the initial setting was made. The temperature shall be 75 ± 3°F at the time operating points readings are taken. Each subsequent operating point shall be within -1.5 mbs and +4 mbs of the corresponding initial operating point.

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4.4.6.3 Transportation vibration. The baroswitch unit shall be secured to a vibration table and vibrated for 90 minutes in each of three mutually perpendicular planes through a frequency range of 10 to 55 Hertz (Hz) at a double amplitude of 0.064 inch. The frequency range shall be uniformly varied from 10 to 55 to 10 Hz making approximately one cycle per minute. Upon completion of this test, the unit shall be visually examined for damage and then tested as specified in 4.4.6.1.1 and 4.4.6.1.2.

4.4.6.4 Shock. The baroswitch unit shall be subjected to one shock of 15 gravity units (g's) on each of its six sides. Each shock impulse shall reach its maximum in 5- to .5 milliseconds and shall have a pulse time of 11 + 1 milliseconds. Upon completion of this test, the unit shall be visually examined for damage and then tested as specified in 4.4.6.1.1 and 4.4.6.1.2.

4.4.6.5 High temperature. The baroswitch unit shall be placed in a test chamber and the chamber temperature raised to and stabilized (see 6.4) at +160 + 5°F. The relative humidity shall be maintained at a maximum of 5 percent. The unit shall be subjected to this temperature and relative humidity for 10 hours. The temperature of the unit shall be reduced without forced cooling to a temperature of 75 + 3°F. After completion of this test, the baroswitch unit shall be tested as specified in 4.4.6.1.1 and 4.4.6.1.2.

4.4.6.6 Low temperature. The baroswitch unit shall be placed in a test chamber and stabilized at a temperature of -65 + 5°F for 24 hours or until the temperature of the unit stabilizes (see 6.4). The temperature of the unit shall then be increased without forced warming to a temperature of 75 + 3°F. After completion of this test, the baroswitch unit shall be tested as specified in 4.4.6.1.1 and 4.4.6.1.2.

4.4.7 Life cycle. The test set-up of Figure 2, or approved equivalent, may be used for this test. Each element shall be run through 250 pressure cycles that shall vary from approximately 116 to 1014 to 116 mbs at a standard rate of pressure change. The element contact shall make and break a 200-milliamperes non-inductive current once during each of the cycles. Upon completion of this test, the baroswitch element shall be tested as specified in 4.4.3 and 4.4.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The baroswitch standard shall be packaged in accordance with dwg 8861441 or 8861442, as applicable, using the specified moisture barrier.

5.1.2 Level C. The baroswitch standard shall be packaged in accordance with dwg 8861441 or 8861442, as applicable, except that the moisture barrier may be omitted.

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5.2 Packing.

5.2.1 Level A and C. The baroswitch standard shall be packed in accordance with dwg 8861441 or 8861442, as applicable.

5.3 Marking. Marking shall be as specified in accordance with dwg 8861441 or 8861442, as applicable.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4). Data specified in 4.3.4 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Definitions.

6.5.1 Unit. The term "unit" as used in this specification designates the complete Baroswitch Standard T-4167A or T-4168A.

6.5.2 Element. The term "element" designates one of the baroswitch elements which comprise the unit.

6.5.3 Contact closure reading. A contact closure reading is defined as the pressure value observed on "A" of Figure 1 at a contact closure indication.

6.5.4 Contact closure indication. Contact closure indication is defined as the change in lamp F of Figure 2 from a dim glow to normal brilliance (determine normal brilliance in advance by closing S1). This change shall occur during a maximum change of 0.13 mbs or less as observed on "A" of Figure 1.

6.5.5 Standard rate of pressure change. A standard rate of pressure change is a rate equivalent to 1000 $\pm$ 200 feet of equivalent pressure altitude per second.

6.5.6 Operating point. The average of any three consecutive contact closure readings of a baroswitch element without changing pressure setting or temperature shall be known as the operating point.

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6.5.7 Repeatability error. The absolute difference between the minimum and maximum contact readings in any set of three consecutive closure readings shall be known as repeatability error.

6.5.8 Temperature error. The difference between 0°F operating point and ambient (75 ± 3°F) operating point shall be known as temperature error. Likewise, the difference between ambient operating point and 130°F operating point shall be known as temperature error.

6.5.9 Pressure exercise. A pressure exercise shall consist of three pressure cycles from approximately 116 to 1014 to 116 mbs at a standard rate of pressure change. (A baroswitch element shall be exercised immediately prior to the next closure sensing dive any time the element is subjected to a fixed pressure for longer than 10 minutes.)

6.5.10 Present tolerances. If the baroswitches are to be present prior to shipment, the following values shall be used:

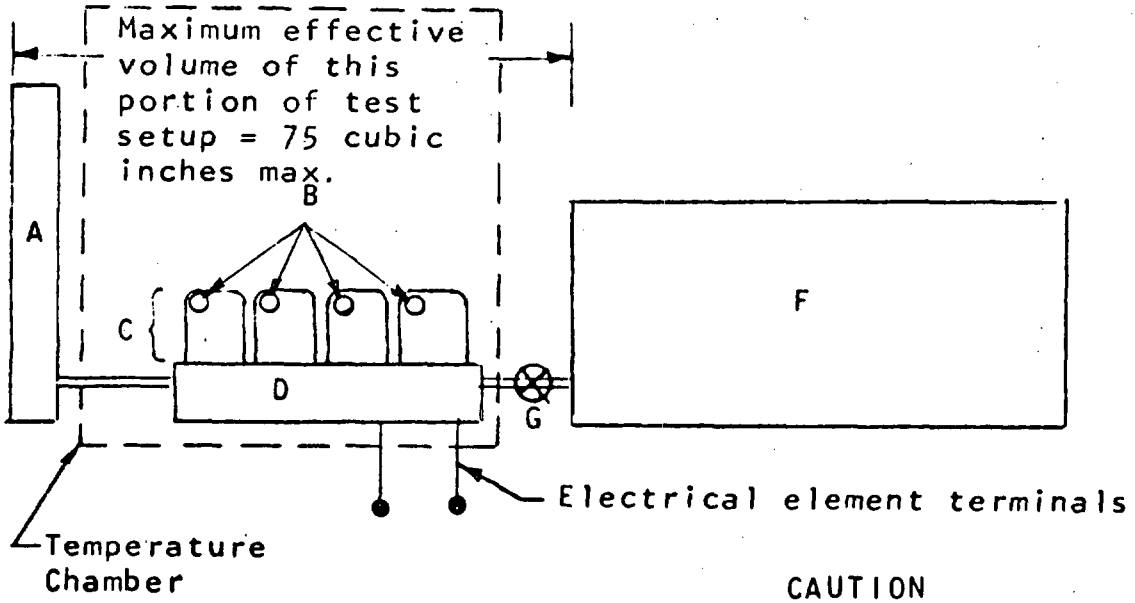
Unit	S1 element	S2 element
First T-4167A or T-4168A/4650	± 103 feet	5350 ± 106 feet
Second T-4167A or T-4168A/650	± 92 feet	1350 ± 94 feet

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1190-A072

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CAUTION

In no event shall a test circuit be used which will produce a current flow in excess of 0.3 ampere through the element contacts.

A - Mercury barometer - precision 0.05 mm Hg - accuracy 0.3 mm Hg.

B - Element adjustment ports for calibration.

C - Four units each - baroswitch standard T-4167A and T-4168A less dust cover.

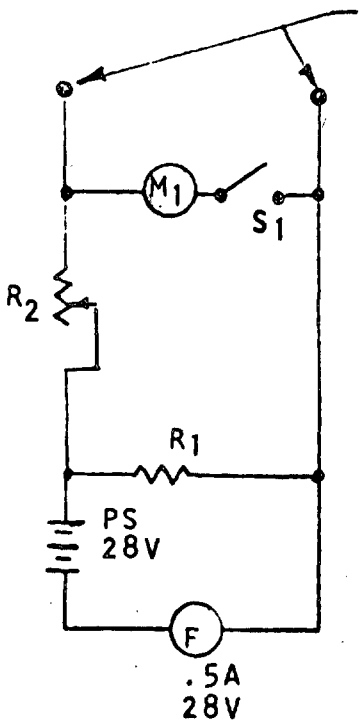
D - Manifold (dwg. C8595991 or equivalent).

E - Pressure producing and regulating system (pressure pump, vacuum pump, manostat). NOTE: Provision for dehydrated air, 40 percent maximum relative humidity.

F - Valve.

Figure 1. Acceptable Pressure Switch Test Circuit And System

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CAUTION

In no event shall a test circuit be used which will produce a current flow in excess of 0.3 ampere through the element contacts.

M_1 - DC Milliammeter, 0-250 MA.

R_1 - Resistor 560 ohms ± 5% 1/2 watt, fixed.

R_2 - Resistor, adjustable to permit 200 ma. current flow through M_2 when S_2 is closed in figure 2.

PS - Power Supply 28 Volts ± 2VDC.

F - Lamp 28 Volts, 500 ma.

S_1 - SPST Switch

Figure 2. Acceptable Life Cycle Test Circuit

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